

Reproducibility Report: Maglio & Polman (2014), Study 1

Minimal version

Urvi Suwal

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Goals

1. Load a published dataset with `here()` and `read_excel()`
2. Reproduce the ANOVA and effect sizes reported in the paper
3. Say clearly what did and did not reproduce, and why

This is the **minimal** version: the structure and the questions are here, and you write the code. If you get stuck, `reproducibility-report-bumper-rails.qmd` has the same steps with most of the code filled in. It is fine to peek at it for one step and then come back.

error: `true` is set at the top of this file, so the document still renders while some chunks are unfinished. The chunks that still fail show their error message in red.

Before you start, read Study 1 in the paper: [original_paper/maglio-polman-2014.pdf](#) (two pages).

The study

Maglio, S. J., & Polman, E. (2014). Spatial orientation shrinks and expands psychological distance. *Psychological Science*, 25(7), 1345–1352. <https://doi.org/10.1177/0956797614530571>

Methods summary. Researchers recruited 202 volunteers at the Bay Street subway station in Toronto. Half were waiting on the westbound platform and half on the eastbound platform. Each participant was randomly assigned to rate how far away one of four stations felt (1 = very close, 7 = very far): Spadina (two stops west), St. George (one stop west), Bloor-Yonge (one stop east), or Sherbourne (two stops east). The authors ran a 2×4 ANOVA on the distance ratings, then compared eastbound and westbound travelers separately at each station.

Target outcomes. This is the result you are trying to reproduce, quoted from the Study 1 results section:

We carried out a 2 (orientation: toward, away from) $\times$ 4 (station: Spadina, St. George, Bloor-Yonge, Sherbourne) analysis of variance (ANOVA) on closeness ratings, which revealed no main effect of orientation, $F < 1$, and a main effect of station, $F(3, 194) = 24.10$, $p < .001$, $p^2 = .27$. This main effect was qualified by the predicted interaction between orientation and station, $F(3, 194) = 16.28$, $p < .001$, $p^2 = .20$. We decomposed this interaction by the subjective-distance ratings between participants traveling east and west for each of the four subway stations. Westbound participants rated the stations to the west of Bay Street as closer than did eastbound participants; this effect was obtained for both the station one stop to the west (St. George, $p < .001$, $p^2 = .28$) and the station two stops to the west (Spadina, $p = .001$, $p^2 = .20$). The opposite pattern held true for stations to the east of Bay Street. Eastbound participants rated the stations to the east of Bay Street as closer than did westbound participants; this effect was obtained for both the station one stop to the east (Bloor-Yonge, $p = .053$, $p^2 = .08$) and the station two stops to the east (Sherbourne, $p < .001$, $p^2 = .24$). Figure 1 summarizes these results.

p^2 is **partial eta squared**, an effect size. You will compute it below.

Step 1: Load packages

`readxl` (reads Excel files) and `broom` (turns model output into tibbles) are installed along with the tidyverse, but `library(tidyverse)` does not load them, so we load them by name.

```
install.packages(c("tidyverse", "here"))
```

```
library(tidyverse)
library(readxl)
library(broom)
library(here)
```

Step 2: Load the data

The authors posted one Excel file per study. We only need Study 1. The columns are described in `docs/codebook.md`.

```
d <- read_excel(here("data", "S1_Subway.xlsx"))

glimpse(d)
```

Rows: 202

Columns: 4

```
$ DIRECTION <chr> "EAST", "EAST", "EAST", "EAST", "EAST", "EAST", "EAST", "EA~
$ DISTANCE  <dbl> 1, 2, 2, 3, 1, 1, 2, 1, 3, 2, 1, 1, 2, 1, 3, 1, 2, 1, 1, 1,~
$ STN_NUMBER <dbl> 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3,~
$ STN_NAME  <chr> "B-Y", "B-Y", "B-Y", "B-Y", "B-Y", "B-Y", "B-Y", "B-
Y", "B--
```

Step 3: Tidy the data

The data are already tidy: one row per participant, one column per variable. But R has read `DIRECTION` and `STN_NAME` in as text. For an ANOVA they need to be **factors**, and we set the order of the station levels to run west to east, so tables and plots come out in map order.

Use `mutate()` and `factor()` to convert both columns. For `STN_NAME`, give `factor()` a `levels` argument listing the stations from west to east.

```
d <-d |> mutate(
  DIRECTION = factor(DIRECTION),
  STN_NAME = factor(STN_NAME, levels = c("SPAD", "STG", "B-Y", "SHER")) # order from the r
)
```

Step 4: Reproduce the analysis

How many people are in each cell?

Count the participants in each direction $\times$ station combination. Does it match the paper (202 total, 101 per platform)? Is every cell the same size?

```
d |> count(DIRECTION, STN_NAME)
```

```
# A tibble: 8 x 3
  DIRECTION STN_NAME      n
  <fct>     <fct>   <int>
1 EAST      SPAD      26
2 EAST      STG       26
3 EAST      B-Y       23
4 EAST      SHER      26
5 WEST      SPAD      25
6 WEST      STG       25
7 WEST      B-Y       26
8 WEST      SHER      25
```

The 2×4 ANOVA

Fit a two-way ANOVA with `aov()`: distance predicted by direction, station, and their interaction. Remember that `y ~ A * B` is shorthand for `y ~ A + B + A:B`.

```
anova_model <- aov(DISTANCE ~ DIRECTION *STN_NAME, data = d)
```

```
# anova_model
summary(anova_model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
DIRECTION	1	0.71	0.713	0.664	0.416
STN_NAME	3	75.16	25.053	23.349	6.01e-13 ***
DIRECTION:STN_NAME	3	52.41	17.471	16.283	1.77e-09 ***
Residuals	194	208.15	1.073		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The numbers in parentheses in “F(3, 194)” are the degrees of freedom: 3 for the effect, 194 for the residuals. Check that yours match.

The teaching team gets:

- Main effect of direction: $F(1, 194) = 0.66$
- Main effect of station: $F(3, 194) = 23.35$
- Interaction between direction and station: $F(3, 194) = 16.28$

Two of those match the paper. One is close but not the same. Keep going; the bonus question at the end asks you to work out why.

Partial eta squared

Partial eta squared is the share of variance an effect accounts for, once the other effects are set aside:

$$\eta_p^2 = \frac{SS_{\text{effect}}}{SS_{\text{effect}} + SS_{\text{residual}}}$$

Compute it for each effect. Hint: `tidy()` from `broom` turns the ANOVA table into a tibble, with the sums of squares in a column called `sumsq`.

```
anova_tib <- tidy(anova_model)

sumsq_residuals <- anova_tib |>
  filter(term == 'Residuals') |>
  pull(sumsq)

anova_tib |>
  filter(term != 'Residuals') |>
  mutate(partial_eta_sq = sumsq/(sumsq+sumsq_residuals))
```

A tibble: 3 x 7

	term	df	sumsq	meansq	statistic	p.value	partial_eta_sq
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	DIRECTION	1	0.713	0.713	0.664	4.16e- 1	0.00341
2	STN_NAME	3	75.2	25.1	23.3	6.01e-13	0.265
3	DIRECTION:STN_NAME	3	52.4	17.5	16.3	1.77e- 9	0.201

Do your values round to the paper’s .27 (station) and .20 (interaction)? Yes!

Station-by-station comparisons

Now compare eastbound and westbound travelers at each station separately. For each station: keep only that station's rows, compute the mean rating in each direction, run a one-way ANOVA of distance on direction, and compute p^2 .

St. George

```
# station rows
STG_d <- d |> filter(STN_NAME == "STG")

# mean ratings
STG_d |>
  group_by(DIRECTION) |>
  summarize(mean = mean(DISTANCE), n = n())
```

```
# A tibble: 2 x 3
  DIRECTION mean    n
  <fct>      <dbl> <int>
1 EAST      2.77    26
2 WEST      1.64    25
```

```
# one way anova
STG_anova_model <- aov(DISTANCE ~ DIRECTION, data=STG_d)

#  $p^2$ 
STG_anova_tib <- tidy(STG_anova_model)

sumsq_residuals <- STG_anova_tib |>
  filter(term == 'Residuals') |>
  pull(sumsq)

STG_anova_tib <- STG_anova_tib |>
  filter(term != 'Residuals') |>
  mutate(partial_eta_sq = sumsq/(sumsq+sumsq_residuals))
```

Spadina

```
# station rows
SPAD_d<- d |> filter(STN_NAME == "SPAD")

# mean ratings
SPAD_d |>
  group_by(DIRECTION) |>
  summarize(mean = mean(DISTANCE), n = n())
```

```
# A tibble: 2 x 3
  DIRECTION mean      n
  <fct>      <dbl> <int>
1 EAST      3.65     26
2 WEST      2.64     25
```

```
# one way anova
SPAD_anova_model <- aov(DISTANCE ~ DIRECTION, data=SPAD_d)

# p2
SPAD_anova_tib <- tidy(SPAD_anova_model)

sumsq_residuals <- SPAD_anova_tib |>
  filter(term == 'Residuals') |>
  pull(sumsq)

SPAD_anova_tib <- SPAD_anova_tib |>
  filter(term != 'Residuals') |>
  mutate(partial_eta_sq = sumsq/(sumsq+sumsq_residuals))
```

Bloor-Yonge

```
# station rows
BY_d<- d |> filter(STN_NAME == "B-Y")

# mean ratings
BY_d |>
  group_by(DIRECTION) |>
  summarize(mean = mean(DISTANCE), n = n())
```

```
# A tibble: 2 x 3
  DIRECTION mean      n
```

	<fct>	<dbl>	<int>
1	EAST	1.61	23
2	WEST	2.19	26

```
# one way anova
BY_anova_model <- aov(DISTANCE ~ DIRECTION, data=BY_d)

# p²
BY_anova_tib <- tidy(BY_anova_model)

sumsq_residuals <- BY_anova_tib |>
  filter(term == 'Residuals') |>
  pull(sumsq)

BY_anova_tib <- BY_anova_tib |>
  filter(term != 'Residuals') |>
  mutate(partial_eta_sq = sumsq/(sumsq+sumsq_residuals))
```

Sherbourne

```
# station rows
SHER_d<- d |> filter(STN_NAME == "SHER")

# mean ratings
SHER_d |>
  group_by(DIRECTION) |>
  summarize(mean = mean(DISTANCE), n = n())
```

	DIRECTION	mean	n
	<fct>	<dbl>	<int>
1	EAST	2.77	26
2	WEST	4	25

```
# one way anova
SHER_anova_model <- aov(DISTANCE ~ DIRECTION, data=SHER_d)

# p²
SHER_anova_tib <- tidy(SHER_anova_model)

sumsq_residuals <- SHER_anova_tib |>
```

```

filter(term == 'Residuals') |>
pull(sumsq)

SHER_anova_tib <- SHER_anova_tib <- SHER_anova_tib |>
  filter(term != 'Residuals') |>
  mutate(partial_eta_sq = sumsq/(sumsq+sumsq_residuals))

```

Fill in this table as you go:

Station	Paper p	Your p	Paper p^2	Your p^2
St. George	< .001	< 0.001	.28	0.28
Spadina	.001	0.001	.20	0.2
Bloor-Yonge	.053	0.053	.08	0.08
Sherbourne	< .001	< 0.001	.24	0.24

Step 5: Reflection

Were you able to reproduce the results you attempted to reproduce? If not, which parts could you not reproduce?

In the paper $F(3,194) = 24.10$ and the one we got was 23.35.

How difficult was it to reproduce the results?

I think it was pretty straightforward and easy to reproduce the results.

What made it hard, and what made it easy? Think about the data files, the variable names, and how much the paper told you about the analysis.

Variable names were straightforward, readable and analysis described in the results were clear.

Bonus questions

1. Why is the station F different? Your F for station (23.35) is close to the paper's (24.10) but not equal, while the interaction matches exactly. Try refitting the model with the predictors in the opposite order, `DISTANCE ~ STN_NAME * DIRECTION`. Does anything change? Look back at your cell counts. What does that suggest about how the authors' software (SPSS) and R's `aov()` each divide up the variance? [docs/anova-in-r.md](#) explains it, but try first.

```
anova_model_2 <- aov(DISTANCE ~ STN_NAME * DIRECTION, data=d)
summary(anova_model_2)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
STN_NAME	3	75.47	25.156	23.446	5.41e-13 ***
DIRECTION	1	0.40	0.402	0.375	0.541
STN_NAME:DIRECTION	3	52.41	17.471	16.283	1.77e-09 ***
Residuals	194	208.15	1.073		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Changing the order of predictors changes the values, but not for the interaction.

2. Orientation or direction? The paper describes the factor as “orientation: toward, away from”. The dataset has DIRECTION: east or west. Are those the same variable? If not, which one did the authors actually put in their ANOVA? Make an **orientation** column with `if_else()` and refit the model with it to check.

not sure if this is what i was supposed to do but this was my understandin of toward and away from the paper’s methods+results.

```
new_d <- d |>
  mutate(
    ORIENTATION = case_when(
      DIRECTION == "WEST" & (STN_NAME == "SHER" | STN_NAME == "B-Y") ~ "toward",
      DIRECTION == "EAST" & (STN_NAME == "SPAD" | STN_NAME == "B-Y") ~ "toward",
      DIRECTION == "EAST" & (STN_NAME == "B-Y" | STN_NAME == "SHER") ~ "away",
      DIRECTION == "WEST" & (STN_NAME == "STG" | STN_NAME == "SPAD") ~ "away",
    )
  )

orientation_model <- aov(DISTANCE ~ ORIENTATION *STN_NAME, data = new_d)

# anova_model
summary(orientation_model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
ORIENTATION	1	12.35	12.35	11.304	0.000955 ***
STN_NAME	3	111.20	37.07	33.935	< 2e-16 ***
ORIENTATION:STN_NAME	1	0.30	0.30	0.275	0.600997
Residuals	170	185.69	1.09		

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
26 observations deleted due to missingness
```

3. Reproduce Figure 1. Plot the mean distance rating for each station, split by direction, with error bars of ± 1 standard error, as in the paper. `stat_summary(fun.data = mean_se)` will draw both the means and the error bars for you.

Going further

Instead of four near-identical chunks, do all four stations in one pipeline. Look up `nest()` and `map()`.